

Exploring MBE Growth Parameters and Material Quality of III-V Topological Insulators Grown on GaSb(111)A Substrates

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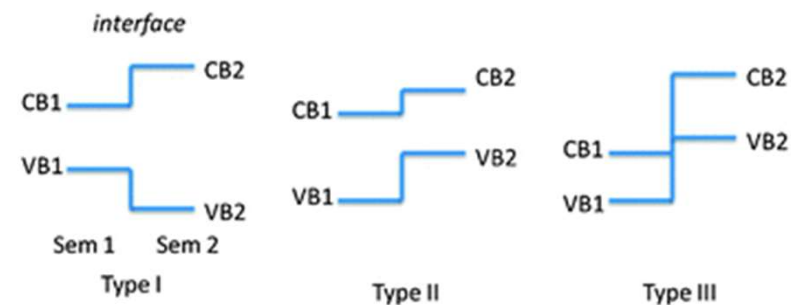
Topological Insulators

- TIs exhibit an insulating bulk and conducting edge states which are robust against scattering
- TIs have largely been constructed from materials that are hard to get or manufacture like Bismuth Selenide or Mercury Telluride
- Finding III-V material system with these characteristics could
 - open the door to TI research to the III-V community at large
 - Ease the integration of these structures with existing technologies
- 2D TIs
 - a manifestation of the quantum spin Hall effect (QSHE) wherein the spin-orbit coupling, acting as an intrinsic electric field, induces spin degenerate Landau Levels
 - Two crucial ingredients for topological gap: band inversion and spin-orbit coupling



Topological Insulators from III-V materials

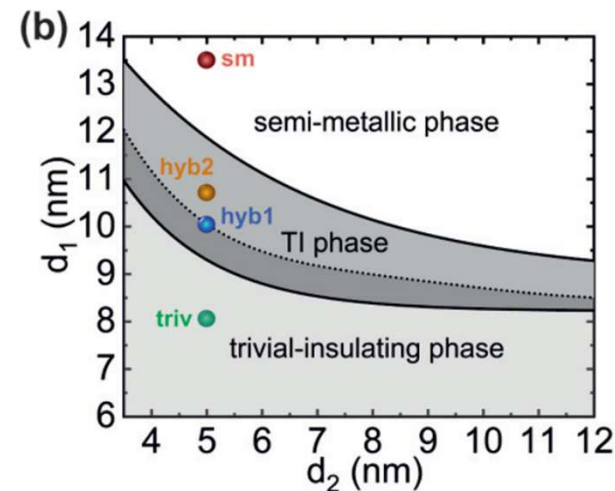
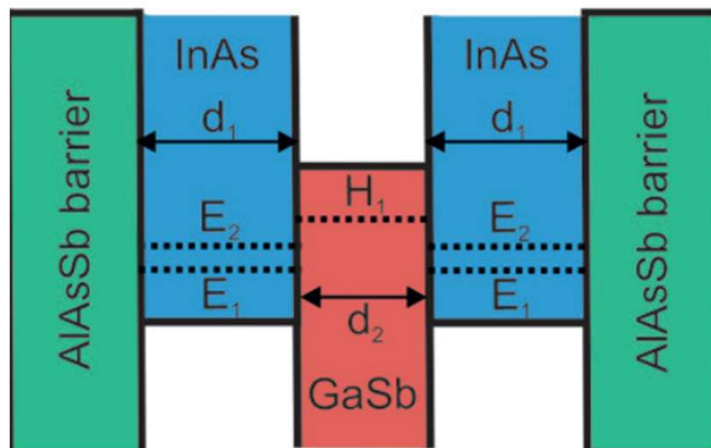
- InAs/GaSb heterostructures provide an ideal playground for TI engineering
 - Strong spin-orbit coupling, broken band gap (Type III), localized hole/electron states, integrability with existing technologies
- The broken band gap allows for a tuning of the topological phase via a shift in the relative alignment of the electron/hole subbands through:
 - quantum well (QW) thickness
 - QW strain to enhance spin-orbit coupling effects
 - the application of a gate voltage to shift the fermi energy [1]



[2]

Background

- This work is a result of a collaboration with Sven Höfling's group at the University of Würzburg
 - Showed that the TI Phase transition can be tuned via the application of a gate voltage
 - to date, this work has focused on materials grown on (100) surface
 - strong spin-orbit coupling in these systems which may be strengthened by growth on the (111) surface



Background

- Largely though the efforts of Kevin Vallejo and Maddy Drake, our group has experience growing strain-driven GaSb based structures on the (111) surface through our labs previous work growing tensile-strained InGaAs QDs
- The combination should help us explore the physics and possible benefits of the higher in-plane symmetry and out-of-plane polarization of the (111) surface orientation

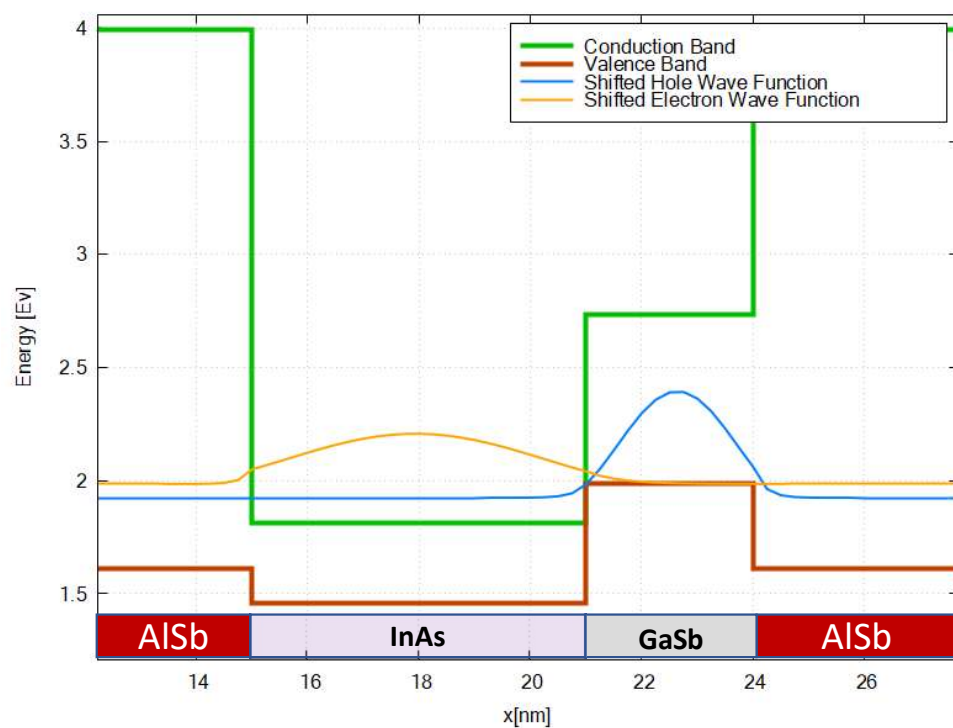
InAs/ Ga₆₈In₃₂Sb (111)A QW



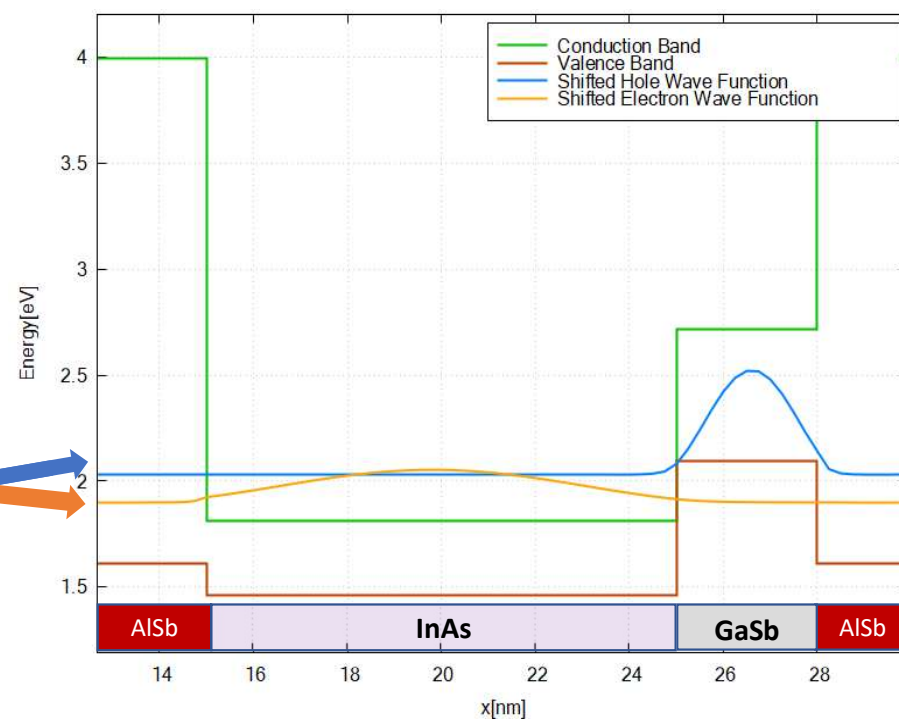
Effects of the QW Thickness

- Alter subband alignment
- Alter orbital overlap
- Simulations run with Nextnano

InAs/GaSb (111)A QW in a Trivial Insulating State



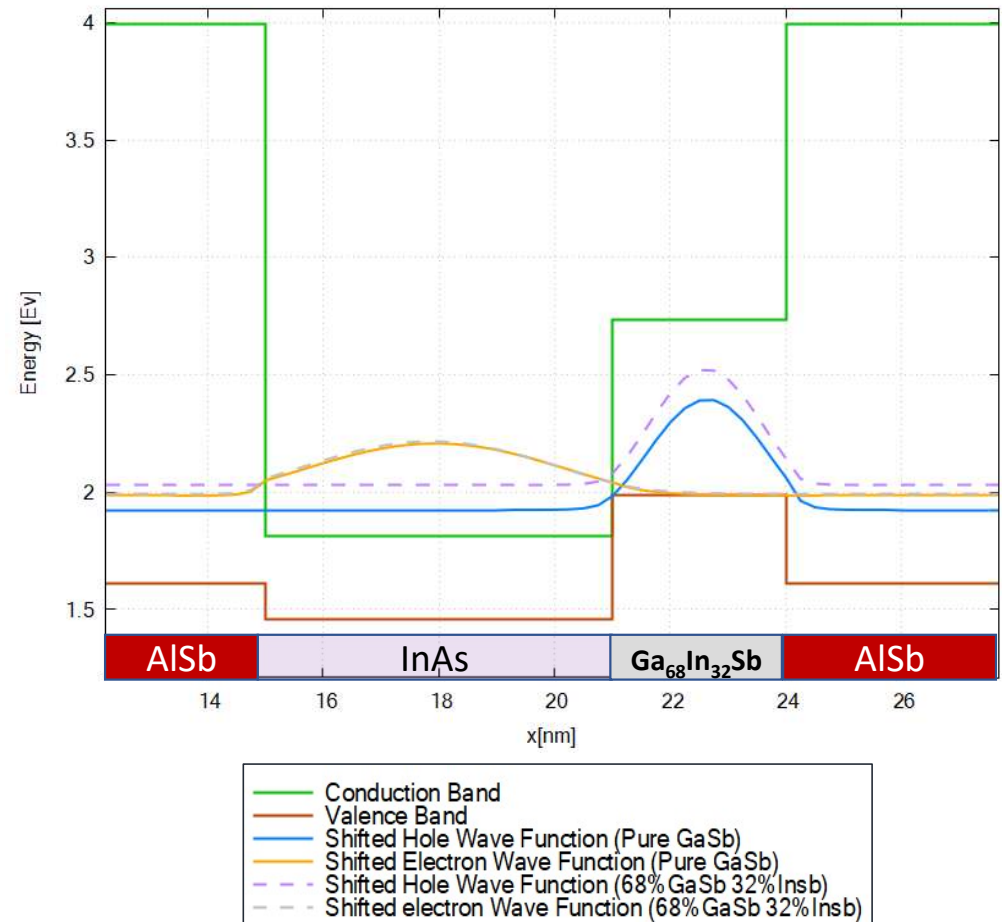
InAs/GaSb (111)A QW in a TI state



Effects of the strain

- Splits the valence band into heavy-hole and light-hole levels
- Pulls the conduction band down below the heavy hole level
- Crossing of the bands happens at thinner well thicknesses which allows for more overlap between hole and electron wave function and an exaggeration of hybridization driven topological band gap[2]

Double Quantum Well with wave functions plotted for Pure GaSb or $\text{Ga}_{68}\text{In}_{32}\text{Sb}$ (dotted)

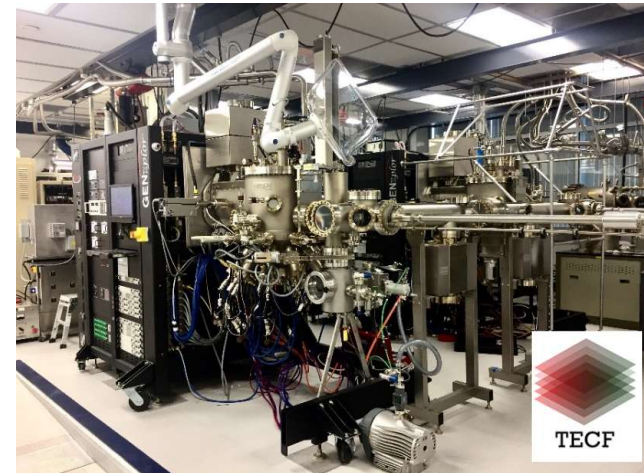


Challenges

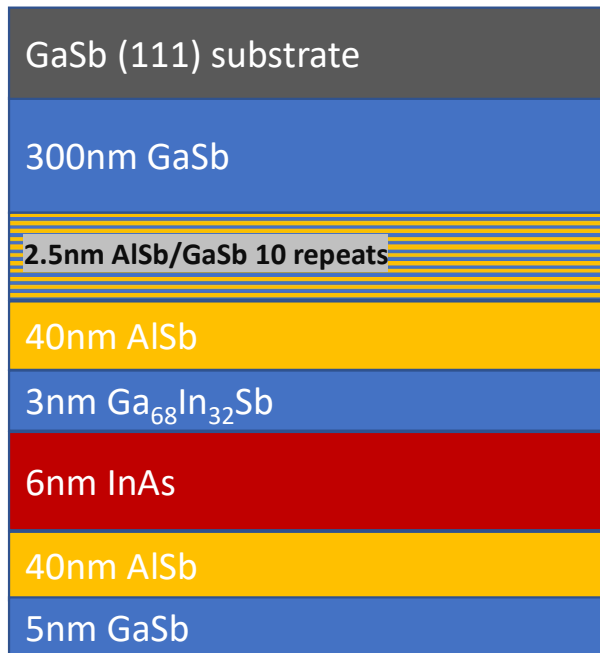
- **Switched from EPI GEN 930 to Veeco GenXplor**
 - New mechanistic geometry meant for a recalibration of growth parameters
- **Switching group V's between quantum wells**
- **Finding ideal growth parameters for (111)**
 - low Sb sticking in the (111)A, V-III Sb BEP ratios must be balanced against feasible growth rates

- The 3x to 5x swap is not an option for anchoring the InAs substrate temperature for (111) surface
- Optimizing substrate and cell temperature ramping between layers

Maintaining precise layer control while minimizing growth rate



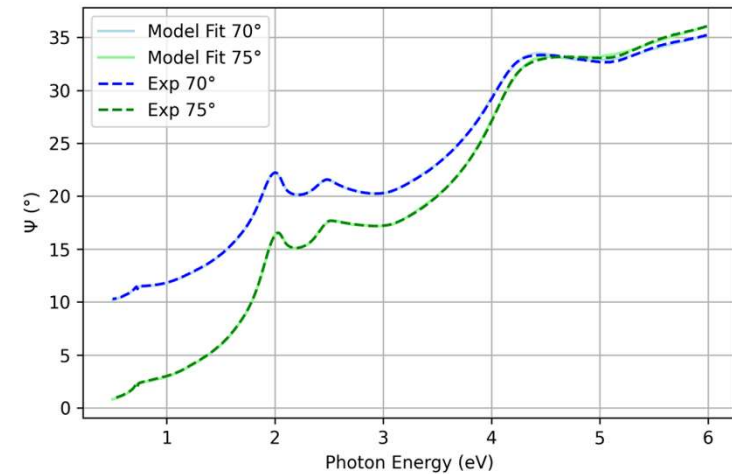
Shutter sequence for Sb to As layer swap



Step Name	Ga shutter	In Shutter	Al shutter	As shutter	Sb shutter
wait					
close Sb					
open arse...					
6nm-InAs					
close As					
wait					
1 ML InSb					
close					

Layer thickness/composition control

- RHEED Oscillations and InAs QDs for slow InAs growth rate (GR)
- Did XRD on a $\text{Ga}_{0.68}\text{In}_{0.32}\text{Sb}$ sample to confirm composition and optical characteristics
- Built an ellipsometry model to confirm layer thicknesses with out expensive TEM
 - The goodness of fit varies, unsurprisingly, with material surface quality



2.3nm Srough

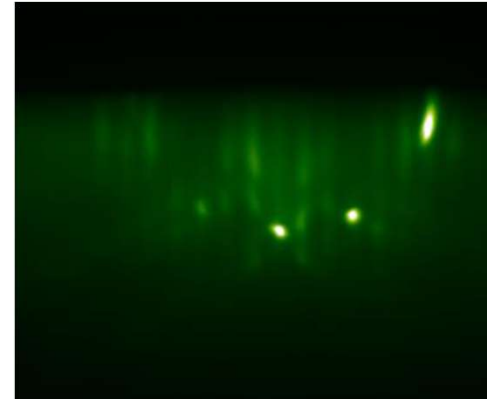
10nm $\text{Ga}_{0.68}\text{In}_{0.32}\text{Sb}$

100 GaSb Substrate

Temperature control

- Indium mount (100) analogues
- Observe 3x to 5x swap of GaSb (100) for bottom end substrate temperature calibration
- Observe deox for top end substrate temperature calibration

3x



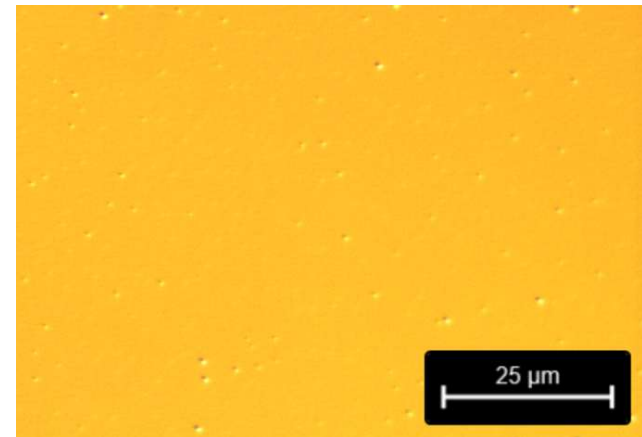
5X



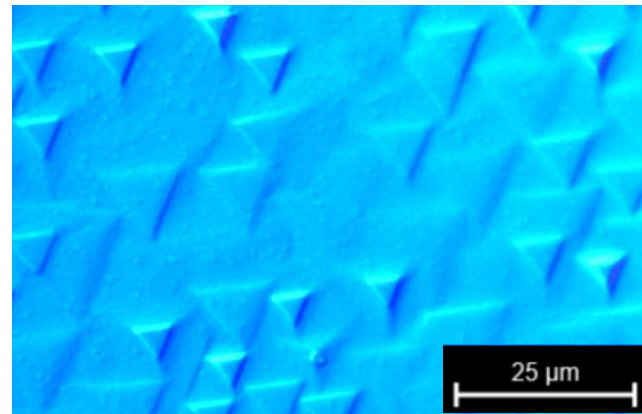
Growing on (111) with parameters that make (100) happy

- Growing at stoichiometric ratio creates a nice surface on (100)
- But results in macroscopic 3D islanding on the (111)
- These results compare well to our labs prior (111) GaSb growths

(100)



(111)



Maximize Sb overpressure for Sb containing layers by reducing layer thicknesses and growth rates

Sample 1

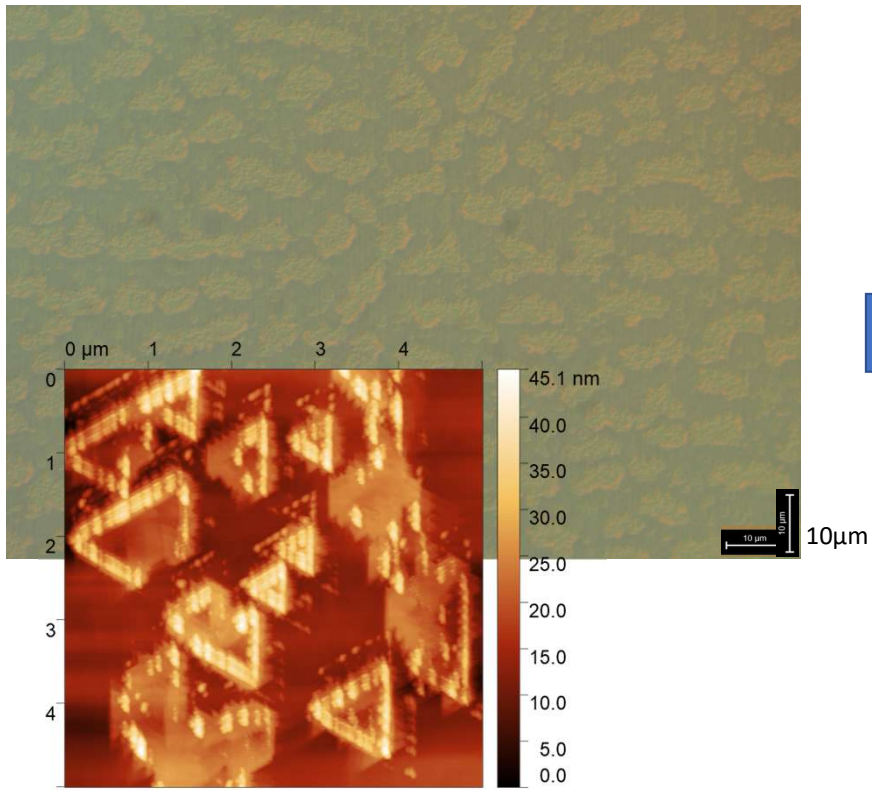
5nm GaSb	
40nm AlSb	V/III flux ratio 10X GR .133ML/s
6nm InAs	
3nm Ga ₆₈ In ₃₂ Sb	V/III flux ratio 7X GR .1ML/s
40nm AlSb	V/III flux ratio 10X GR .133ML/s
2.5nm AlSb/GaSb 10 repeats	
300nm GaSb	V/III flux ratio 28X GR .068ML/s
GaSb (111) substrate	

Sample 2

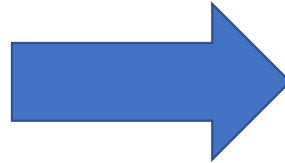
5nm GaSb	
22.5nm AlSb	V/III flux ratio 65X GR .0175ML/s
6nm InAs	
3nm Ga ₆₈ In ₃₂ Sb	V/III flux ratio 30X GR .0348ML/s
22.5nm AlSb	V/III flux ratio 65X GR .0175ML/s
2.5nm AlSb/GaSb 10 repeats	
100nm GaSb	V/III flux ratio 65X GR .024ML/s
GaSb (111) substrate	

Surface quality of Sample 1 vs Sample 2

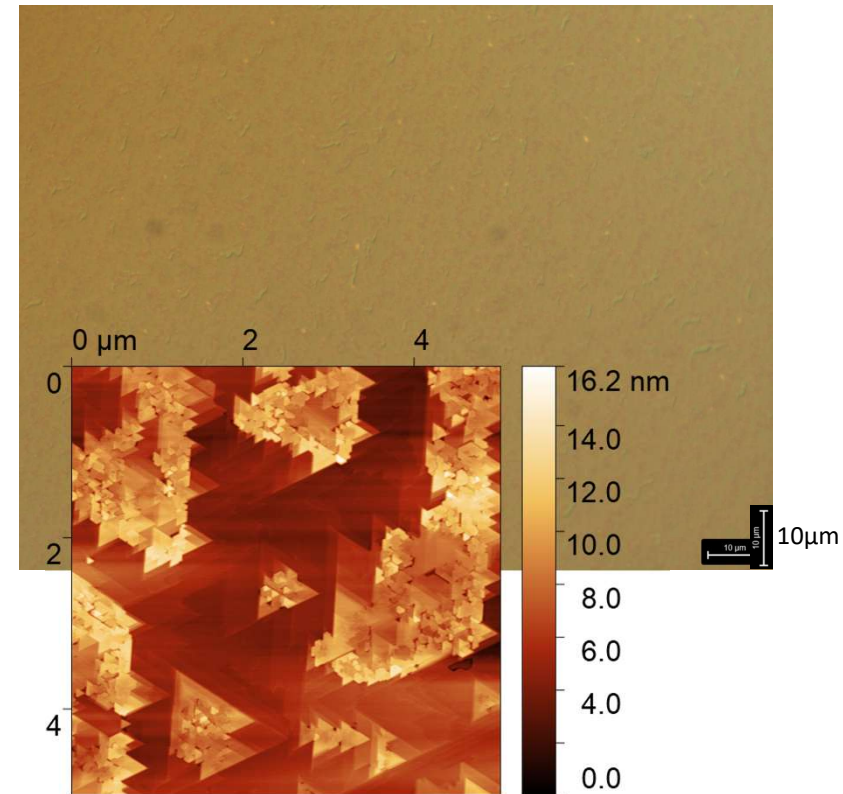
Sample 1



RMS roughness = 6.64nm

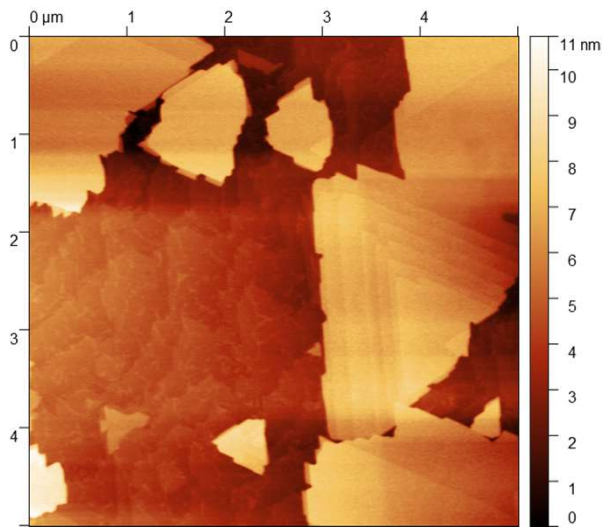


Sample 2

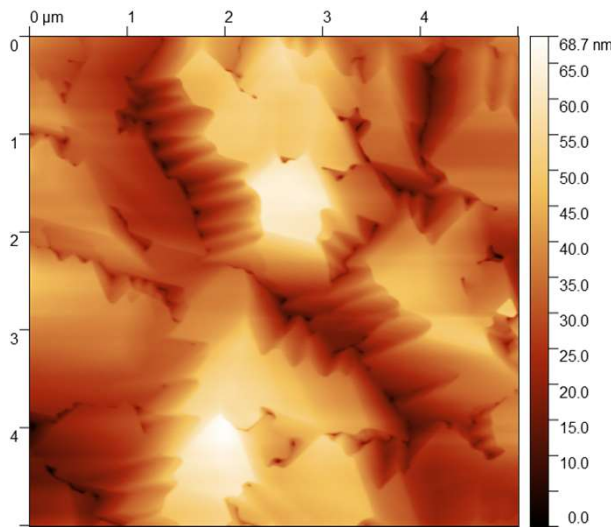
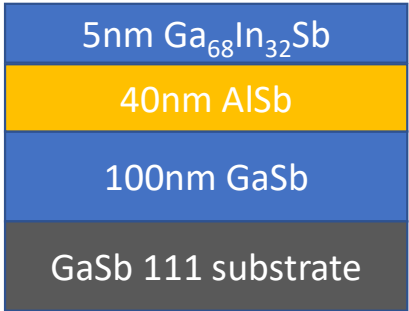


RMS roughness = 2.37nm

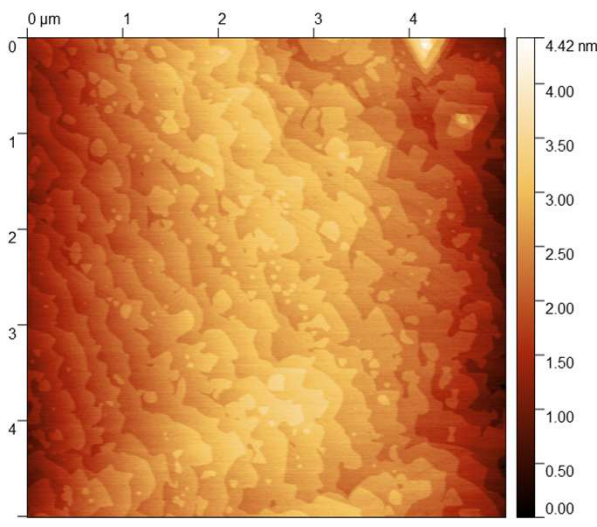
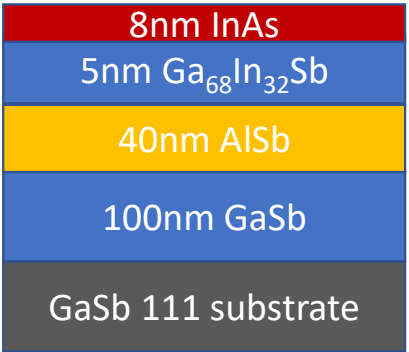
Layer-by-layer growths



RMS roughness = 1.562nm



RMS roughness = 11.4nm

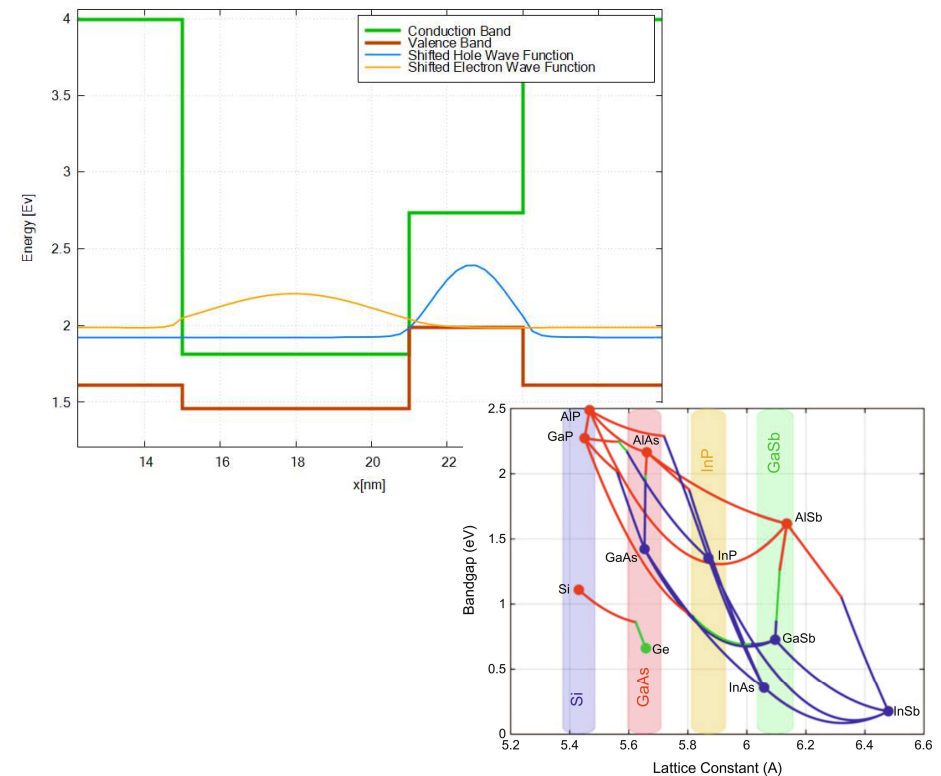


RMS roughness = 0.628nm

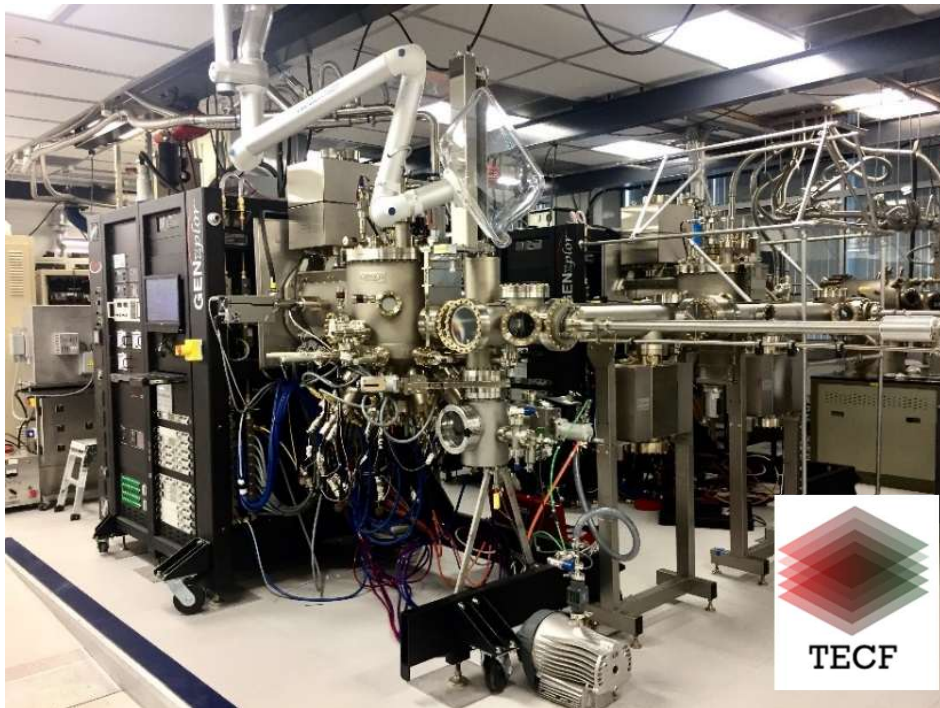
Future Work

- Further optimize growth conditions for isolated InAs and $\text{Ga}_{68}\text{In}_{32}\text{Sb}$ wells
 - Particular investigation of the interface between wells
- TEM to feed back into our ellipsometry models, Magnetotransport measurements, and GR calculations
- Further exaggerate the V/III ratio for Sb containing layers to see if the qualitative trend continues
- In silico experimentation to maximize strain, minimize layer thickness to ultimately maximize the topological band gap

- Magnetotransport measurements to be performed by Prof. Dr. Sven Höfling's group at the University of Würzburg



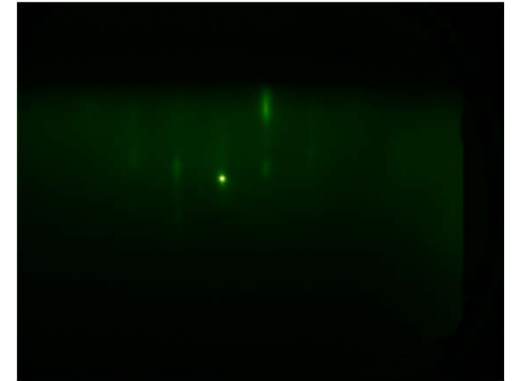
Acknowledgements



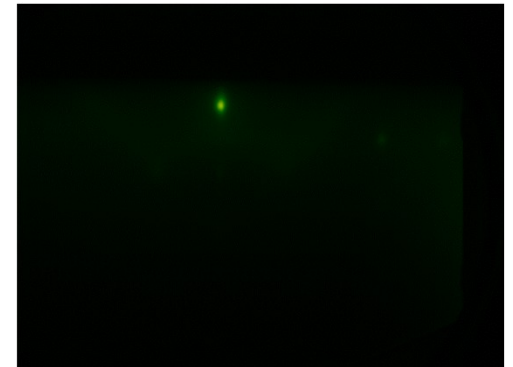
james.rushing@tufts.edu

Tried to grow InAs at a higher temp to simplify and automate shutter control

Ga₆₈In₃₂Sb



InAs

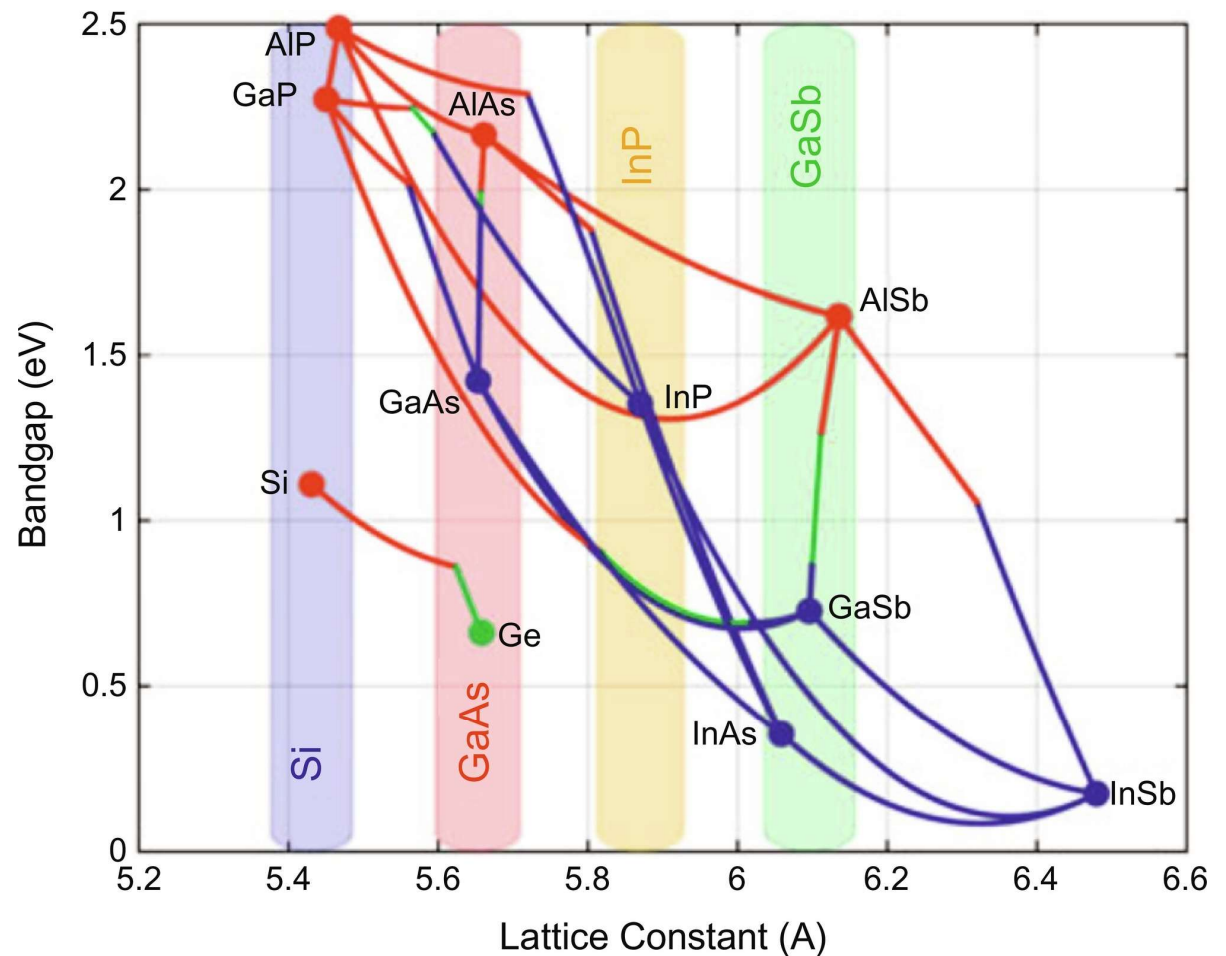


- Sample hazy to the eye and orange peely under Nomarski
- RHEED went spotty Immediately upon deposition of InAs
- For this structure, the substrate temperature is the limiting quality factor for InAs possible to its low bandgap

The Materials

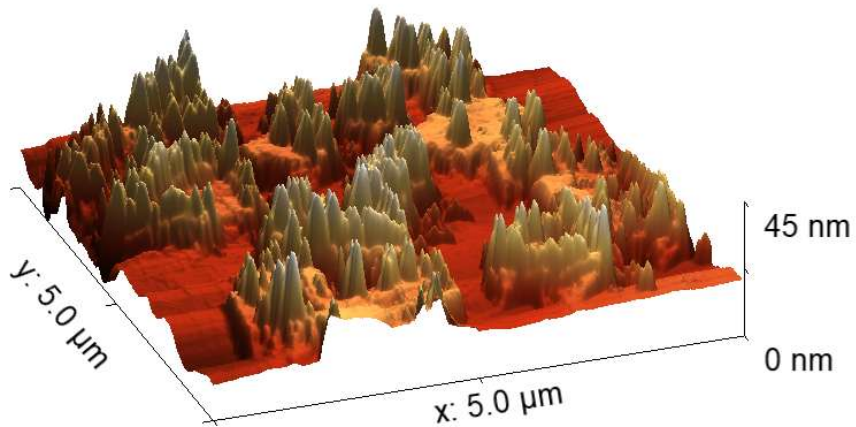
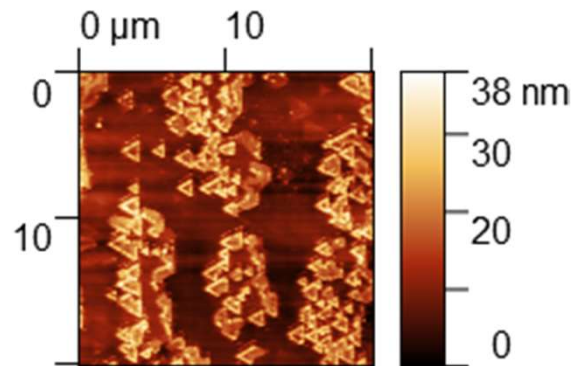
Our systems are in the 6.1 angstrom family, shown in green

- The QSHE is engineered primarily through the interplay between QW thicknesses and strain
- Strain can be tuned via composition along the ternary composition lines

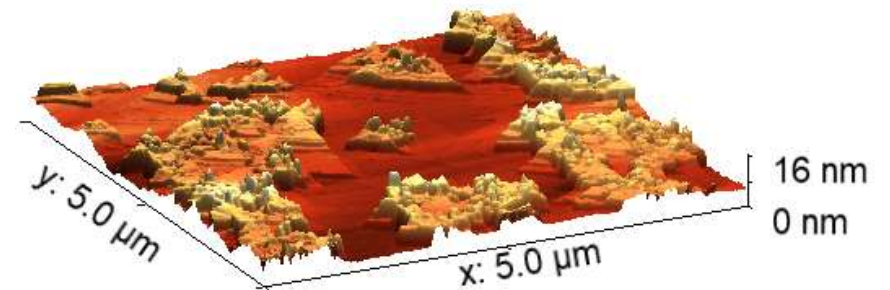
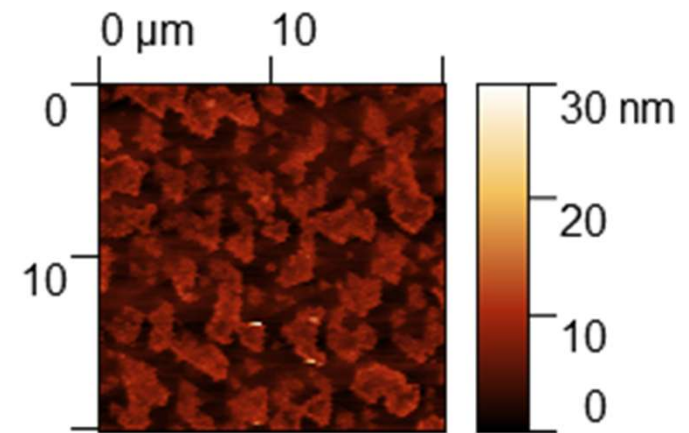


Extra data

Sample 1

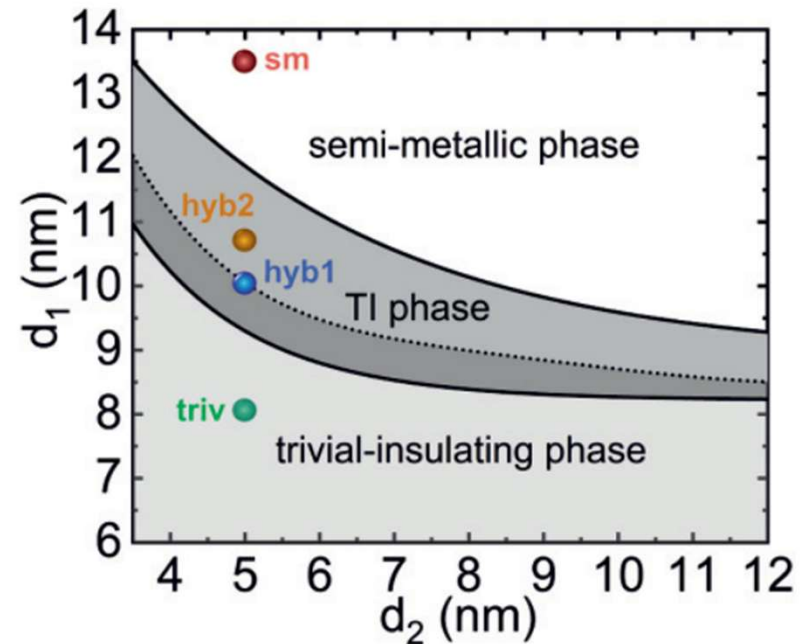
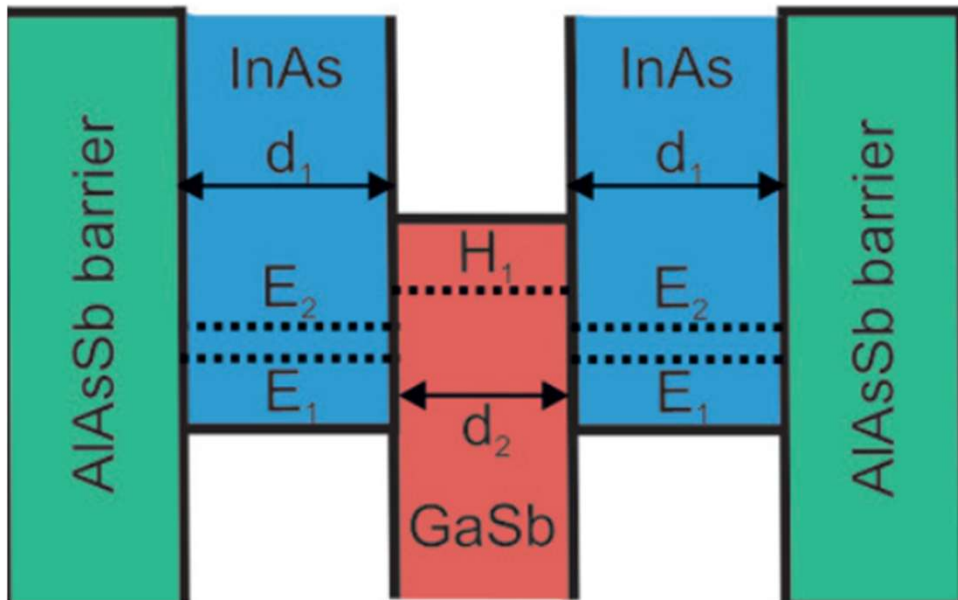


Sample 2



Effects of layer thickness

- Layer thicknesses push hole and electron levels up or down
- Nanometer differences in thickness of either GaSb or InAs layers can tune the material between different trivial and topological phases



PS1

Slide 21

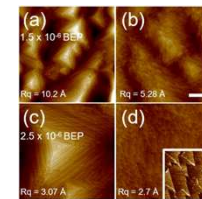
RJ5 Rushing, James, 7/10/2024

PS1 Might not be important but the band structure diagram on p2 shows a double QW structure. You might want to make the point here that people are exploring both TQWs and DQWs, with reasons why one might be preferred to the other.

Paul Simmonds, 7/15/2024

Extra info

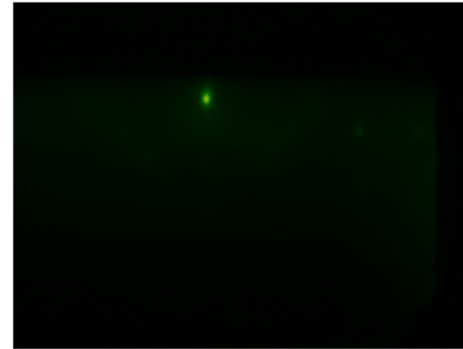
- Are they TIs? Magnetotransport measurements to be performed by collaborators at the University of Wurzburg. The 111 ordination also complicates all fabrication procedures as it does growing.



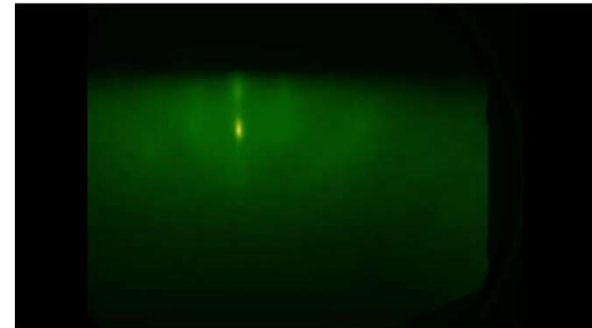
InAs RHEED by Tsub

Probably going to scrap this slide. Too hard to compare RHEED patterns with temp because we used too many different mounting techniques and sometimes our RHEED was just dim

Tsub 510 0324A



Tsub 450 0125a



Tsub 485 0224